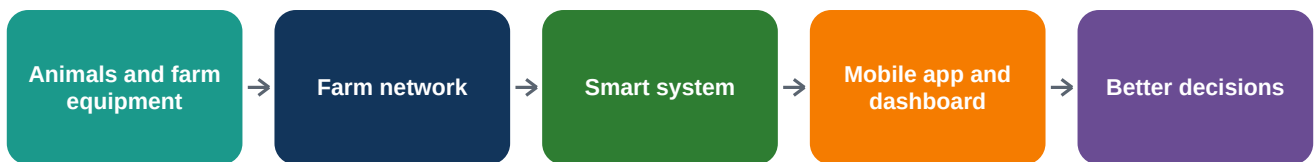


NON-DEVELOPER EXPLANATION GUIDE

Smart Livestock Management System

A plain-language explanation of what the system does, how it works, who uses it, and how it helps farmers manage livestock better.



Purpose of this document

This document explains the livestock management system without software jargon. It is written for farmers, farm managers, students, investors, supervisors, project evaluators, and decision makers who need to understand how the system works and why it is useful.

Prepared as a stakeholder-friendly explanation document

1. The system in simple terms

A quick overview for non-technical readers

The smart livestock management system is like a **digital farm assistant**. It helps the farmer know what is happening with the animals, the water, the feed, the health records, the breeding cycle, the farm workers, and the business side of the farm.

Instead of relying only on memory, notebooks, guesswork, or physical inspection, the system keeps records automatically and gives useful reminders and alerts. It helps the farmer take action earlier, reduce losses, improve productivity, and manage the farm like a modern business.

What it does

Keeps animal records, monitors health, tracks breeding, manages feed and water, records treatments, watches movement, creates alerts, and produces reports.

Who uses it

Farm owners, managers, workers, veterinarians, buyers, inspectors, and service providers can each use the part of the system that applies to them.

Why it matters

It reduces missed vaccinations, late treatment, theft risk, feed waste, poor breeding records, and weak financial control.

How it helps

It turns farm activities into clear information: what happened, where it happened, which animal is affected, and what should be done next.

In one sentence

The system gives every animal a digital identity, collects information from the farm, checks for problems, alerts the right person, and keeps a complete history of farm activities.

2. The problem the system solves

Why livestock farms need more than paper records and manual checking

Livestock farming involves many moving parts. Animals must be fed, watered, vaccinated, treated, bred, weighed, protected, and eventually sold or transferred. On small farms, the farmer may remember most details. On larger farms, this becomes difficult and mistakes become expensive.

Common problems include:

- Animals miss vaccinations or deworming dates because reminders are not automatic.
- A sick animal is noticed late because early signs are subtle.
- Breeding dates, pregnancy checks, and expected birth dates are forgotten.
- Feed and water problems are discovered after animals are already affected.
- Stock theft is noticed late, especially at night or in remote grazing areas.
- Weight gain and profitability are difficult to measure accurately.
- Farm workers complete tasks, but managers have no clear record of what was done.
- Financial records are separated from animal records, so profit per animal is unclear.

Old way

Paper notebooks, manual checking, memory-based reminders, delayed reporting, and decisions based on incomplete information.

Smart system way

Digital records, automatic reminders, sensor alerts, clear dashboards, animal histories, and action-based reports.

The system does not replace the farmer. It helps the farmer see more, remember more, and act faster.

3. How everything connects

The whole system from farm equipment to farmer action

The system works by connecting the real farm to a digital platform. Animals, equipment, sensors, farm workers, and managers all provide information. The platform organizes the information and sends useful alerts or reports back to the right people.

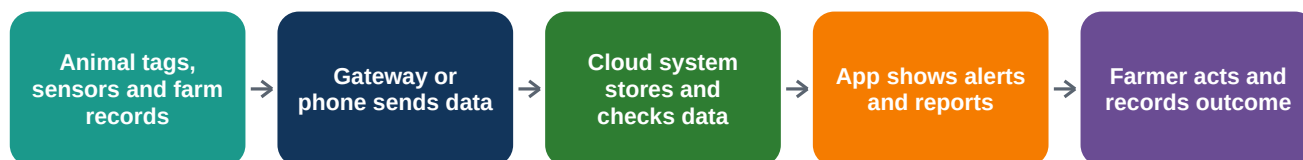


Figure: The basic journey of information through the system.

A simple example is a water tank. A water sensor measures the level. If the level becomes low, the system receives the reading, checks it against the safe level, creates an alert, sends a message to the farmer or worker, and stores the event for future reports.

Farm side

Animals, workers, water tanks, feeding points, gates, paddocks, scales, cameras, and weather conditions.

Connection side

The phone, farm gateway, SIM card, Wi-Fi, LoRa radio network, or other communication method that sends information.

System side

The online platform that stores information, checks rules, prepares alerts, and keeps records safe.

User side

Mobile app, web dashboard, SMS, WhatsApp, email, and printed or PDF reports.

4. Who uses the system

Different people see different information

The same system can be used by several people, but each person should only see the information they need. This keeps the system simple and protects sensitive farm information.

User	What they use it for	Example
Farm owner	Sees the full farm picture: herd size, finances, alerts, production, risk, and reports.	Checks which animals are profitable and which need attention.
Farm manager	Plans daily tasks, checks worker progress, monitors alerts, and reviews animal performance.	Assigns a worker to inspect a sick animal.
Farm worker	Receives tasks and records daily actions in the mobile app.	Marks a water tank refill or vaccination as completed.
Veterinarian	Views health history, treatments, symptoms, medicine records, and follow-up dates.	Adds treatment details and withdrawal period.
Buyer or abattoir	May view approved traceability records for animals being sold.	Checks age, vaccinations, and treatment history.
Inspector or auditor	Reviews compliance reports, movement records, and health records when permitted.	Checks vaccination and treatment documentation.

Important principle

A worker should not see confidential financial information. A buyer should not see all farm operations. Each person receives a role, and the role controls what they can view or change.

5. Every animal gets a digital identity

The foundation of the whole system

The system begins by giving every animal a unique identity. This can be an RFID ear tag, QR tag, printed ear tag number, or a GPS device number. Once the animal is registered, all future records are linked to that animal.

Physical identity

The animal has a visible tag, electronic tag, collar, brand, or device that helps people identify it on the farm.

Digital identity

The system stores a profile for the animal: species, breed, sex, age, weight, health status, breeding status, photos, history, and ownership details.

What an animal profile should contain

- Animal number or tag number
- Species, breed, sex, colour, age, and photos
- Parent information, if known
- Current group, paddock, or location
- Weight history and growth trend
- Vaccination, deworming, and treatment history
- Breeding status and expected birth dates
- Sale status, purchase cost, and financial records
- Notes from workers, managers, or veterinarians

This creates a complete life story for each animal. Instead of records being scattered, everything is stored in one place.

6. The daily farm cycle

How the system fits into normal farm work

The system should support the normal routine of the farm. It should not make farm work more complicated. It should make daily decisions clearer and faster.

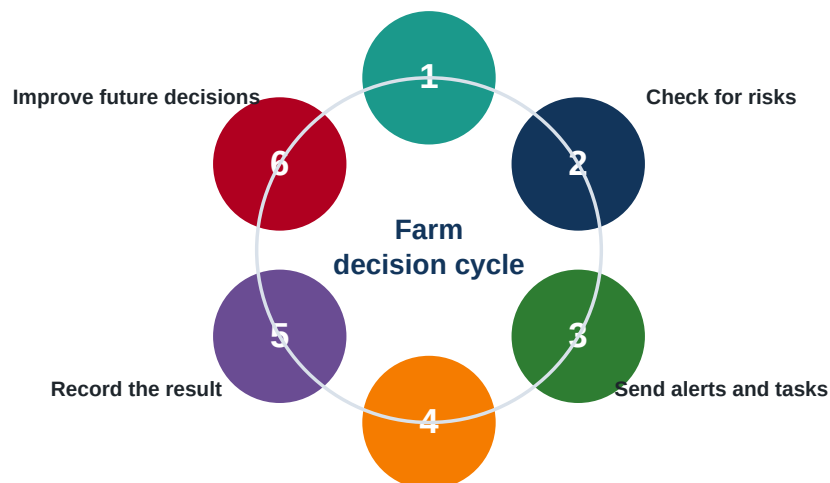


Figure: The daily decision cycle. The system turns farm events into action and learning.

In practice, a farmer may open the app in the morning and see the most important items first: sick animals, low water, feed warnings, overdue vaccinations, animals due for breeding, and security events. The farmer does not need to search through every record before acting.

7. Main parts of the system

The system is made of practical farm modules

A module is simply a section of the system that handles one important farm activity. The modules work together so that information from one part can help another part.

Animal records

Stores each animal profile and full life history.

Health

Records sickness, symptoms, medicine, vet visits, recovery, and deaths.

Vaccination

Manages vaccine dates, reminders, proof, and overdue alerts.

Breeding

Tracks heat, mating, pregnancy, expected birth, and offspring.

Feed

Tracks feed stock, feeding plans, usage, cost, and shortage alerts.

Water

Monitors water levels, pump activity, leaks, and water quality.

Weight and growth

Tracks weight gain, underperforming animals, and market readiness.

Security

Uses location, geofences, gate sensors, and cameras to reduce theft risk.

Finance

Links costs and income to animals, groups, and the whole farm.

Reports

Creates simple summaries for owners, vets, buyers, inspectors, or funders.

8. Health management in plain language

Finding problems earlier and keeping a health history

The health section helps the farmer know which animals need attention and what has been done for them before. It can use both human observations and automatic sensor information.

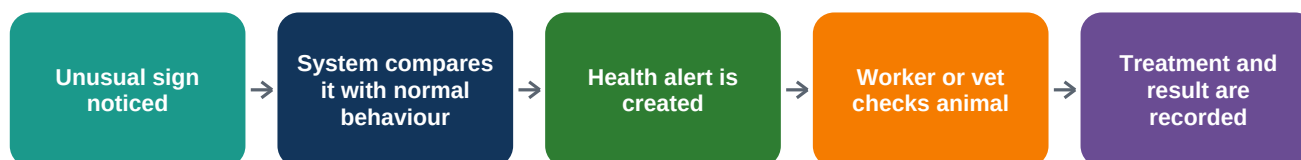
Manual health information

A worker, manager, or veterinarian records symptoms, diagnosis, medicine, dosage, treatment date, follow-up date, and recovery status.

Automatic health signs

Sensors or cameras may notice reduced movement, unusual temperature, abnormal drinking, poor feeding, coughing sounds, isolation, or limping.

How a health alert should work



Example: If a cow usually moves normally but suddenly becomes inactive, drinks less water, and has a high temperature reading, the system should warn the farmer that the animal may be sick. The system should not pretend to be a veterinarian. It should say that the animal needs inspection.

Best practice

Every health alert should allow the farmer to mark the result as confirmed, false alarm, under observation, treated, or referred to a veterinarian. This helps the farm keep better records and improves future recommendations.

9. Vaccination, treatment, and withdrawal periods

Preventing missed dates and unsafe sales

This part of the system acts like a health calendar and medicine record book. It reminds the farm when animals are due for vaccination, deworming, follow-up treatment, or pregnancy checks.

Record	Why it matters	What the system should do
Vaccination	Prevents disease and supports compliance.	Remind the farmer before the due date and mark overdue animals clearly.
Deworming	Improves animal health and weight gain.	Track date given and next expected date.
Medicine treatment	Keeps a clear history of illness and treatment.	Record medicine, dosage, vet, date, and outcome.
Withdrawal period	Protects food safety after medicine use.	Warn if milk or meat should not be sold yet.
Follow-up	Ensures the animal is checked again after treatment.	Create a task for the worker or vet.

Plain example

If an animal receives medicine that requires a 7-day withdrawal period, the system should clearly show that milk or meat from that animal must not be sold until the safe date passes.

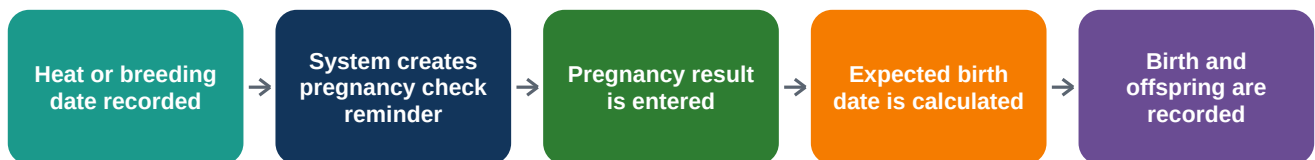
10. Breeding and reproduction

Helping the farm improve births, fertility, and planning

Breeding management is one of the most valuable parts of a livestock system because missed heat cycles, forgotten mating dates, and poor pregnancy tracking can reduce farm income.

The system should help the farmer track:

- Animals ready for breeding
- Heat signs or suspected heat alerts
- Mating or artificial insemination date
- Male animal or semen source used
- Pregnancy check date
- Expected birth date
- Birth result and offspring records
- Animals with repeated breeding failure



A farmer should be able to open the app and quickly see which animals are open, pregnant, near birth, lactating, or due for breeding.

11. Feed and water management

Avoiding shortages, waste, and animal stress

Feed and water directly affect animal health, growth, milk production, fertility, and profit. The system should help the farm know whether enough feed and water are available, where shortages are happening, and how much the farm is spending.

Feed management

Tracks feed stock, daily usage, animal group feeding, feed cost, expected days remaining, and low-stock alerts.

Water management

Tracks tank levels, trough levels, flow rate, pump status, leaks, drinking patterns, and water quality where sensors are available.

How the system should behave

- Warn before feed runs out, not after the store is empty.
- Show which paddock, barn, or group is affected by low water.
- Detect unusual drinking patterns that may indicate sickness, heat stress, or water system failure.
- Connect feeding records with weight gain and production so the farmer can see whether feed is producing results.
- Show feed cost as part of the farm's profit calculation.

Plain example

If a feed bin has enough feed for only three more days, the system should warn the manager early so they can order more feed before animals are affected.

12. Weight, growth, and production

Knowing which animals are performing well

The system should help the farmer measure progress. For meat animals, progress may mean weight gain. For dairy animals, it may mean milk yield. For breeding animals, it may mean fertility and healthy offspring.

Farm question	What the system should show
Is this animal growing properly?	Weight history, average daily gain, and comparison with expected growth.
When should this animal be sold?	Estimated date when it reaches target weight or market condition.
Which animals are underperforming?	Animals gaining too little weight or producing less than expected.
Is feed being used efficiently?	Feed cost compared with weight gain or milk production.
Which animals are most valuable?	Production history, breeding value, sale value, and profit estimate.

Plain example

If two animals eat similar feed but one gains much less weight, the system should highlight the weaker performer. The farmer can then check for parasites, sickness, poor feed access, or genetic weakness.

13. Land, environment, and security

Managing open spaces, housing conditions, and theft risk

The system can also help manage the farm environment. This includes grazing areas, water points, barns, kraals, gates, and farm boundaries.

Grazing and pasture

Shows paddocks, grazing history, recovery periods, overgrazing risk, and water point usage.

Housing environment

Monitors temperature, humidity, ventilation, ammonia, dust, lighting, and heat stress risk where sensors are installed.

Security

Uses geofences, gates, location alerts, cameras, and night movement checks to reduce theft risk.

Farm map

Shows paddocks, boundaries, roads, gates, water points, feed points, and animal groups on a simple map.

Security example

If an animal leaves the farm boundary at night, the system should immediately send a high-priority alert. If a gate sensor also shows that a gate opened at the same time, the system can treat the event as more serious.

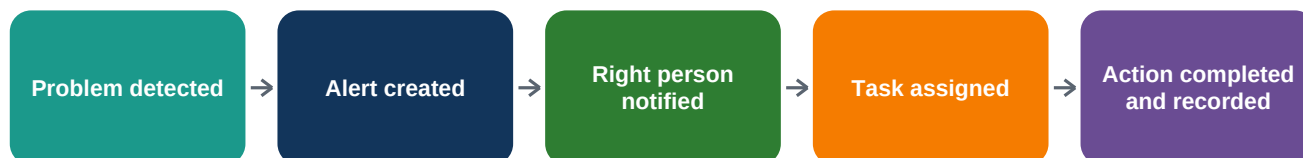
Environment example

If a poultry house, piggery, or dairy barn becomes too hot or poorly ventilated, the system should warn the farm before the animals are stressed or production drops.

14. Alerts and tasks

The system must turn information into action

The system is only useful if it helps people act. That is why alerts and tasks are central. An alert tells the user that something needs attention. A task tells someone what to do about it.



Alert level	Meaning	Example
Low	Informational or early reminder.	Vaccination due in 7 days.
Medium	Needs attention soon.	Feed stock may run out in 4 days.
High	Important problem that should be handled quickly.	Animal is showing possible sickness signs.
Critical	Urgent risk requiring immediate action.	Animal left geofence at night or water is empty.

Good alert design

An alert should clearly say what happened, where it happened, which animal or area is affected, how serious it is, and what action is recommended.

15. The role of AI

AI should support decisions, not replace people

Artificial intelligence, or AI, simply means the system can look for patterns in the data. It can compare today's behaviour with normal behaviour and warn the farmer when something looks unusual.

AI should be explained to users in simple language. Instead of saying 'model anomaly score exceeded threshold', the system should say 'This animal is moving much less than usual and may need inspection.'

AI feature	Plain-language meaning	Useful result
Health risk	Checks whether an animal's behaviour looks unusual.	Early warning before disease becomes severe.
Heat detection	Looks for signs that a female animal may be ready for breeding.	Better timing for mating or artificial insemination.
Growth prediction	Estimates how fast an animal is gaining weight.	Shows when the animal may be ready for market.
Feed forecast	Looks at how much feed is used and how much remains.	Warns before feed runs out.
Theft risk	Combines location, time, gate activity, and movement patterns.	Raises stronger alerts for suspicious movement.

Important warning

AI should never be presented as a final medical diagnosis. It should support the farmer and veterinarian by identifying animals or farm areas that need attention.

16. What happens behind the scenes

A non-technical view of the backend

Behind the mobile app and dashboard, there is a central system that receives information, organizes it, checks it, stores it, and sends messages. For a non-developer, it can be understood as four simple parts.

1. Digital filing cabinet

Stores animal records, farm records, treatment history, breeding records, feed records, water records, and financial records.

2. Farm message receiver

Receives data from sensors, phones, scales, tags, cameras, and gateways.

3. Farm rule checker

Checks whether something is overdue, too low, too high, missing, abnormal, or risky.

4. Notification sender

Sends alerts through the app, SMS, WhatsApp, email, or dashboard notifications.

Example of behind-the-scenes logic

- A sensor sends a water level reading.
- The system checks which tank the sensor belongs to.
- The reading is stored with the date and time.
- The system checks whether the level is safe.
- If the level is too low, an alert is created.
- The farmer or worker receives the alert.
- After the tank is refilled, the action is recorded.

17. What information the system stores

Explained without database jargon

The system should store information in organized categories. This makes it easy to find the history of an animal, the condition of a paddock, or the performance of the farm.

Information category	Examples of what is stored
People	Owners, managers, workers, vets, contacts, and user permissions.
Farms	Farm name, location, boundaries, paddocks, water points, buildings, and equipment.
Animals	Identity, breed, age, sex, photos, status, history, and ownership details.
Health	Symptoms, treatments, medicines, vet visits, recovery, deaths, and withdrawal periods.
Breeding	Heat records, mating dates, pregnancy status, expected birth dates, and offspring.
Production	Weights, growth, milk records, feed conversion, and market readiness.
Resources	Feed stock, water levels, pump records, and environmental conditions.
Security	Movement alerts, geofence events, gate openings, and camera evidence.
Money	Costs, income, sale prices, feed costs, medicine costs, and profit reports.
Reports	Summaries, export files, compliance documents, and farm performance records.

Simple rule

Every important farm action should leave a record. This record helps the farmer explain what happened, learn from it, and make better decisions later.

18. Mobile app and dashboard

What the user should see

The mobile app should be simple enough for daily farm use. The web dashboard can be more detailed for managers, owners, and supervisors.

Mobile app

Best for field work: scan animal tags, receive alerts, complete tasks, add treatments, record births, add weights, take photos, and work offline.

Web dashboard

Best for management: view reports, analyse trends, review finances, manage workers, check devices, and export documents.

Main mobile app screens

- Home dashboard with urgent alerts first
- Animal list and animal profile pages
- Scan RFID or QR tag screen
- Add health, breeding, weight, birth, sale, or death record
- Tasks assigned to workers
- Farm map and paddock view
- Feed and water status
- Reports and summaries
- Offline sync status

The first screen should answer: What needs attention today?

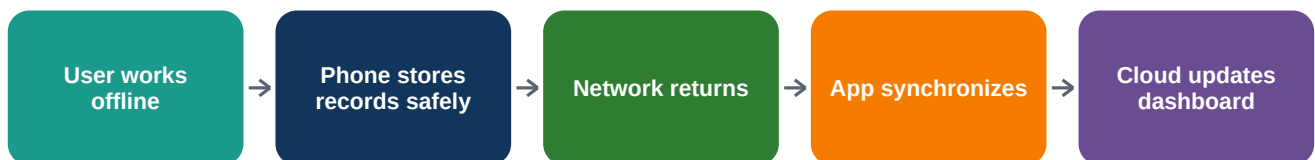
19. Offline use and rural conditions

The system must work even with weak network coverage

Many farms have poor or unreliable internet. The system should still be useful when the phone has no signal. This is especially important for rural areas.

When offline, the app should still allow users to:

- View already downloaded animal records
- Scan an animal tag or QR code
- Add health, breeding, birth, death, weight, and task records
- Take photos and attach them to records
- Complete worker tasks
- Prepare reports or notes for later upload



Plain explanation

Offline mode means the phone keeps temporary records until the internet returns. When connection is restored, the app sends those records to the main system.

20. Reports and business value

Turning records into useful decisions

Reports help the farm owner understand what is happening over time. They also help with veterinary visits, buyer confidence, compliance, insurance, loans, and investment presentations.

Herd report

Shows total animals, births, deaths, sales, health status, and groups.

Health report

Shows sickness cases, treatments, recovery, deaths, and overdue follow-ups.

Breeding report

Shows pregnant animals, expected births, missed cycles, and fertility performance.

Feed and water report

Shows feed usage, water usage, shortages, pump failures, and costs.

Production report

Shows weight gain, milk production, market readiness, and underperforming animals.

Financial report

Shows income, expenses, profit per animal, profit per group, and major cost drivers.

Decision example

If the financial report shows that feed costs are rising but weight gain is low, the farmer can investigate feed quality, parasites, illness, breed performance, or feeding competition.

21. How the system should be introduced

Start simple, then add advanced technology

The system should not be introduced all at once. A step-by-step rollout is safer, cheaper, and easier for users to adopt.

Phase	What to introduce	Main benefit
Phase 1: Digital records	Animal profiles, health records, vaccination reminders, breeding records, weight records, basic reports.	Creates reliable farm records and improves organization.
Phase 2: Farm alerts	Feed and water alerts, task management, offline mobile app, SMS or WhatsApp notifications.	Helps the farm act on problems quickly.
Phase 3: Devices and sensors	RFID readers, weighing scales, water sensors, feed sensors, GPS devices, environmental sensors.	Reduces manual checking and improves real-time awareness.
Phase 4: Intelligence	AI health alerts, heat detection, growth prediction, feed forecasting, theft risk scoring.	Supports better decisions and earlier intervention.
Phase 5: Advanced expansion	Cameras, drones, traceability, buyer portal, insurance reports, advanced analytics.	Turns the system into a full livestock business platform.

Recommended starting point

Start with animal identity, health records, vaccination reminders, breeding records, weight records, feed and water records, worker tasks, and reports. Add expensive sensors only after the basic system is working well.

22. Benefits for different stakeholders

Why each group would care

Stakeholder	Main benefits
Smallholder farmer	Better reminders, fewer lost records, early problem alerts, and clearer animal histories.
Commercial farmer	Productivity tracking, worker control, cost management, traceability, and management reports.
Veterinarian	Better health history, treatment follow-up, medicine records, and disease trend visibility.
Farm worker	Clear tasks, easier reporting, less confusion, and proof of completed work.
Buyer	More confidence in animal history, treatment status, age, and traceability.
Investor or funder	Clearer farm performance, risk control, and measurable improvement over time.
Government or inspector	Better compliance records, disease monitoring support, and movement documentation.

Core value

The system improves visibility, accountability, animal welfare, productivity, and business control.

23. Example: A normal day using the system

How the pieces work together in practice

The following example shows how the system should work on a real farm day.

Time	What happens	How the system helps
Morning	The manager opens the dashboard.	The system shows urgent alerts first: two animals need inspection, one water tank is low, and five calves are due for vaccination.
Morning	A worker receives tasks on the app.	The worker sees which animals to inspect and which paddock needs water.
Midday	The worker scans an animal tag and records symptoms.	The system attaches the observation to the correct animal history.
Afternoon	The vet treats one animal.	Medicine, dosage, follow-up date, and withdrawal period are recorded.
Afternoon	A water tank is refilled.	The worker marks the task complete and the alert closes.
Evening	A gate opens unexpectedly.	The system checks time, location, and animal movement, then sends a security alert if suspicious.
End of day	The owner reviews the summary.	The system shows completed tasks, unresolved alerts, new records, and tomorrow's priorities.

This is the main purpose of the system: not just storing data, but helping people act at the right time.

24. What a good system must do

Non-technical requirements checklist

A strong livestock management system should be practical, reliable, affordable, and easy to use. It should match real farm conditions, not only ideal technology conditions.

Requirement	What it means in simple terms
Easy to use	A farmer or worker should learn the main actions quickly without needing programming knowledge.
Works offline	The app should still record data when internet is unavailable.
Clear alerts	Alerts should explain the problem, location, seriousness, and recommended action.
Role-based access	Different users should only see the information they need.
Reliable records	The system should not lose animal histories, health records, or financial data.
Flexible setup	Small farms can start with tags and mobile app; larger farms can add sensors and AI.
Affordable expansion	The farm should add advanced equipment gradually as value is proven.
Data protection	Farm data should be private, backed up, and protected from unauthorized access.
Useful reports	Reports should help with real decisions, not just look impressive.

25. Simple glossary

Plain meanings of common terms

Term	Plain meaning
RFID	An electronic tag that can be scanned to identify an animal.
GPS	A location system used to show where an animal or device is.
Sensor	A small device that measures something, such as water level, temperature, movement, or weight.
Gateway	A farm device that collects sensor information and sends it to the internet.
Dashboard	A screen that shows the farm summary, alerts, charts, and reports.
Alert	A warning that something needs attention.
Task	A job assigned to a person, such as checking an animal or refilling water.
Geofence	A digital boundary on a map. The system warns if an animal leaves the area.
Offline mode	The app can still work without internet and sync later.
AI	A system feature that looks for patterns and unusual behaviour to support decisions.
Traceability	The ability to show an animal's history from birth or purchase to sale.
Withdrawal period	The safety waiting period after medicine before milk or meat can be sold.

26. Final summary

What the system should achieve

The smart livestock management system should help farms move from scattered records and reactive decisions to organized, informed, and timely management.

At its best, the system should:

- Give every animal a complete digital history.
- Help the farmer notice sickness, poor growth, missed breeding opportunities, and shortages earlier.
- Remind the farm about vaccines, treatments, pregnancy checks, follow-ups, and tasks.
- Connect animal performance with feed, water, health, breeding, and financial records.
- Protect animals through location alerts, geofences, gate monitoring, and security records.
- Support better teamwork between owners, managers, workers, and veterinarians.
- Produce reports that help with business decisions, compliance, sales, and planning.
- Work in real rural conditions, including weak network coverage and limited power.

The main message

This system is not just a tracking tool. It is a full farm management assistant that helps people care for animals, reduce losses, improve productivity, and make better business decisions.

End of non-developer explanation guide